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Letter to the Editor

Comparative study of methods of the determination of Kearns parameter in zirconium

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ABSTRACT

Various methods of Kearns "*f*" parameter evaluation were compared for their consistency and dependency on measurement cross section of the sample and variation in the microstructure across different cross sections. The study showed that, EBSD (Electron Back Scattered Diffraction) method is more consistent in comparison to X-ray based techniques for the "*f*" parameter determination especially in case of recrystallized microstructures.

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1. Introduction

Along with more conventional representations of grain/phase size and shape, crystallographic texture also constitutes an integral part of microstructural description and of structure–property correlation [1–3]. While the former can be described and interpreted relatively easily, description and interpretation of texture is rather an involved subject [1–3]. Typically, texture in a polycrystalline material is represented through Orientation Distribution Function (ODF) [1–3]. This is, however, far more complex than simple scalar quantities, such as an average grain size, and hence may complicate straight-forward structure–property correlations – especially in materials of low crystal symmetry. This is one of the reasons for the apparent popularity of the Kearns parameter (also known as "*f*" parameter): as a simplified index of crystallographic texture for Zirconium (Zr) based components as used by the nuclear industry [4,5]. In principle "*f*" parameter represents resolution and summation of the single crystal properties for a given crystal direction from each of the constituent crystals along the direction of interest. Since many of the properties of the Zr (thermal expansion, creep etc.) are related to how the basal poles of the grains are aligned with respect to the sample reference frame, Kerans *f* parameter is invariably used to represent the fraction of basal poles aligned along a given direction [4]. Past studies have shown good correlation between "*f*" and properties of Zr components [6–15]. This had prompted routine use of "*f*" parameter as a quality specification for Zr-components used in the nuclear reactors.

In the original method of Kearns [4], intensities of XRD (X-ray diffraction) lines, along different cross sections of the sample, were used to estimate the "*f*". As this translates into relative contributions of XRD peaks on a typical inverse pole figure (IPF) representation, the original method of Kearns [4] will be referred, in this study, as IPF method. This relatively easy to use method can be used without a texture cradle or goniometer. "*f*" parameter can be also derived from basal pole figures (PF) along different cross-sections [15,16], and also through appropriate ODF calculations. The IPF and PF methods depend on the availability of the appropriate peaks. In the ODF method, arguably, the missing peaks can be accounted for through alternate peaks & subsequent PF inversions and use of numerical tools, e.g. series-expansion or binning [2,3]. Even then, ODF calculations have the limitations of XRD, in terms of detecting minority texture components [3], and possible numerical errors, e.g. positivity correction, etc. [2,3]. The other alternative is to estimate "*f*" from microtexture data. With faster EBSD cameras, this method can now provide adequate statistics at an excellent accuracy [3].

The routine use of "*f*" parameter, especially by the nuclear industry, and the fact that "*f*" is expected to have excellent correlation with many properties [4–6], puts emphasize on its accurate measurement. However, with the availability of different methods for its determination, it is highly desirable and interesting to have a comparison of these methods for their sensitivity (if any) to various experimental variables like measurement cross-section and variation in microstructure across different cross sections. Previously an elaborate exercise for finding out the variation of the "*f*" using PF method form different laboratories was reported [16]. The focus of that study was to bring out variations due to sample preparation and measurement method of each participating laboratory. Present study, on the other hand, tries to make a comprehensive comparison of different methods of "*f*" parameter determination across various microstructures and cross sections. The objective is to find the most consistent and/or suitable method for "*f*" parameter determination.

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2. Measurement of *f* parameter

Four methods (IPF, PF, ODF, and EBSD) of "*f*" determination were explored. All the three principal cross sections, i.e. RD–TD, RD–ND, ND–TD (see Fig. 1 for the description) were used. Three Zr samples (Rex-Zry-4, Zry-2 Rod and Hot-ex-Zr-2.5Nb) with different prior processing and correspondingly different microstructure, see Table 1, were used. Use of these three samples, with different microstructures, was aimed at exploring possible variations or scatter in "*f*" parameter measurement with variation in microstructure. "Rex-Zry-4" represents a sample with fully recrystallized and homogenous microstructure across all cross sections. "Zry-2 Rod" sample was similar in microstructure to "Rex-Zry-4" but exhibits axial texture. Hot-ex-Zr-2.5Nb on the other hand exhibits microstructural variation across different cross sections. This set of samples thus represents three widely different microstructural conditions and textures and serve the objective of bringing out the role of measurement cross section and variation in microstructure on the "*f*" parameter measurement.

A Panalytical MRD system was used for all XRD measurements, while an EDX-TSL EBSD system on a Fei Quanta 3d FEG SEM (scanning electron microscope) was used for EBSD scans. To maintain reasonable statistics, a relatively large area scan (2 mm × 2 mm in case of recrystallized samples) for EBSD and oscillations (±5 mm) were used for all XRD measurements. 3 μm step size was used for the EBSD scans of recrystallized samples with 97% success in automated indexing. For the 'Hot-ex-Zr-2.5Nb', however, a finer step size of 100 nm was used and the estimated success in automated EBSD indexing was about 70%.

For the IPF method, $2-\theta$ scans were taken at a step size of 0.001° and at 2 s per step of acquisition time. This plus the use of X-ray lens and multi-channel area detector ascertained good X-ray counts. For the PF method (0 0 0 2) pole figures were measured, while ODFs were estimated from the inversion of 4 incomplete pole figures. These were (0 0 0 2), $(0\,1\,\bar{1}\,2)$, $(0\,1\,\bar{1}\,3)$ an $(0\,2\,\bar{2}\,2)$ for all three cross-sections. For ND–TD section, however, (0 0 0 2) peak had low intensities and hence an additional $(1\,0\,\bar{1}\,0)$ was measured. The PF measurements were done up to a tilt angle of 85°. Although the X-ray set up used in the present study employs state of the art X-ray optics, defocusing becomes significant beyond 80° and hence the data beyond 75° was neglected for analysis. Raw PF data was corrected by using pole figure measurement on powder samples made from the samples used in the present study (which served as pseudo random standards).

Procedures to estimate "*f*" by IPF and PF methods are described respectively in [4,16]. ODFs were calculated and basal volume fractions were estimated using open source software "Mtex". "*f*" was then calculated, in the respective cross-sections, as,

$$f_i = \sum V_i \cos^2 \phi_i \tag{1}$$

where f_i is the *f* parameter for a given cross-section along direction '*i*' ('*i*' being RD, TD or ND), V_i is the volume fraction of the (0 0 0 1) basal fiber at an angle ϕ_i with '*i*'. For EBSD, volume fractions, for respective fiber orientation, were taken from direct estimates of appropriate area fractions.

Table 1
Samples and processing conditions used in the present study.

Sample description	Code	Condition	Remarks
Recrystallized zircaloy-4	Rex-Zry-4	Cold worked to 50% by rolling followed by recrystallization at 750 °C for 2 h	The microstructure is fully recrystallized and completely homogeneous across all principal cross sections of the sample
Recrystallized zircaloy-2 rod sample	Zry-2 Rod	Processed by extrusion and a series of pilgering + annealing steps	Used as end plug material. The processing conditions enforce axial symmetry in textural distribution in this material
Hot extruded Zr-2.5Nb	Hot-ex-Zr-2.5Nb	Deformed by extrusion at 800 °C	The microstructure is a typical lamellar structure of elongated α grains separated by thin β stringers. Microstructure is very different across different cross sections of the sample. This sample can be considered as partially recrystallized.

3. Results and discussion

Fig. 2 shows the microstructures along the RD–TD cross section of the three samples considered for the study. It is clear that samples "Rex-Zry-4" and "Zry-2 Rod" show recrystallized and nearly equiaxed α grains. The microstructures of these samples along other principal cross sections were also quite similar and hence are not shown. "Hot-ex-Zr-2.5Nb" sample, however,

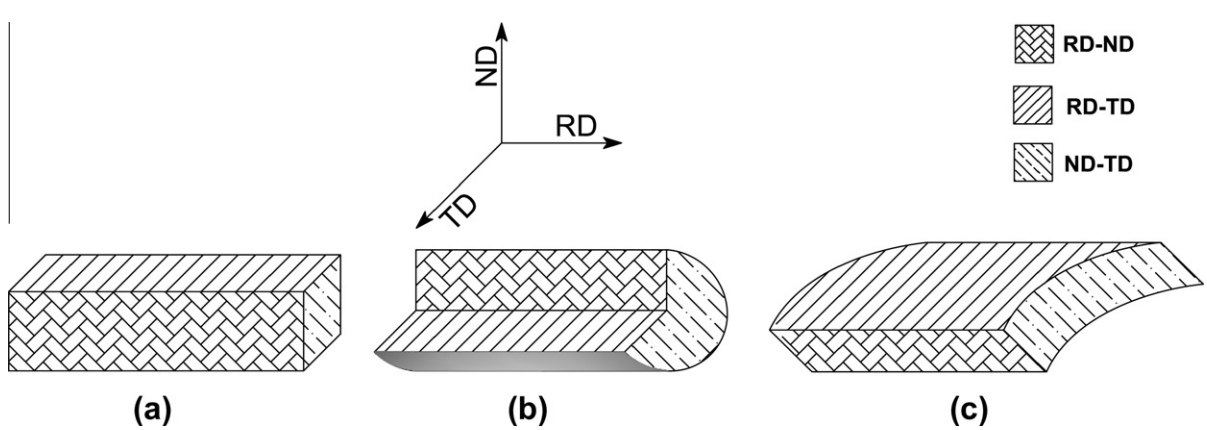

Fig. 1. Schematic defining the reference directions and planes for the texture measurements. RD, TD, and ND refer to reference, transverse and normal directions respectively. (a), (b) and (c) are respectively for "Rex-Zry-4", "Zry-2 Rod" and "Hot-ex-Zr-2.5Nb".

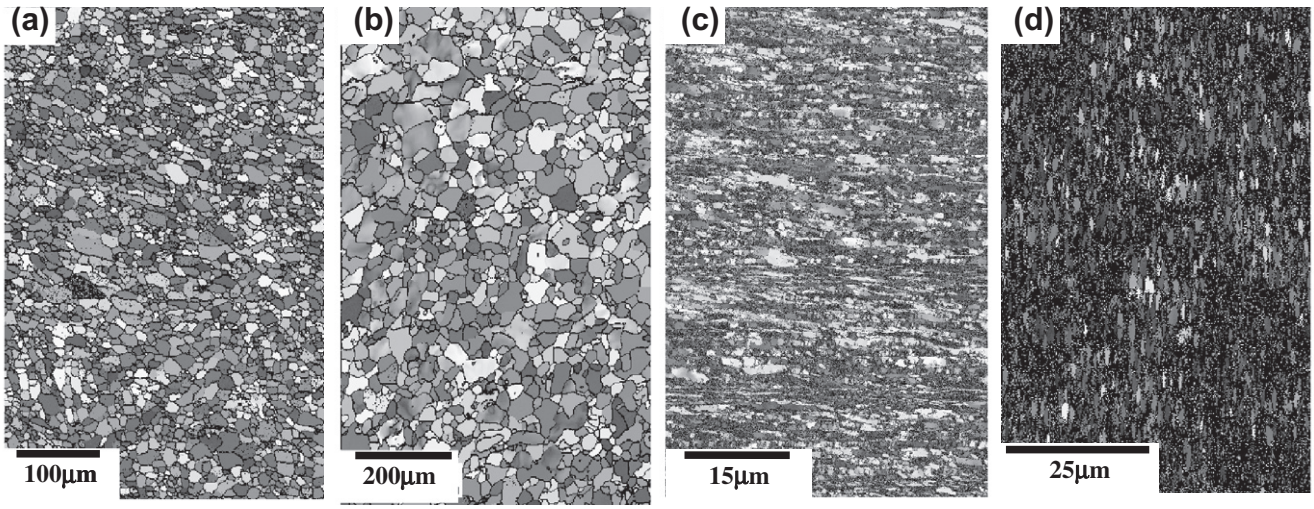

Fig. 2. EBSD microstructures along the RD–TD cross section of the three samples considered for the study: (a), (b) and (c) are respectively for "Rex-Zry-4", "Zry-2 Rod" and "Hot-ex-Zr-2.5Nb" and (d) microstructure along the ND–TD cross section of "Hot-ex-Zr-2.5Nb".

exhibited fine and elongated grains of α in RD–TD cross section as a result of hot extrusion. As expected, the microstructure of this sample in the ND–TD cross section was quite different, Fig. 2d.

Fig. 3 shows the consolidated results of the texture determination by different techniques for the "Rex-Zry-4" sample. Since this sample had similar microstructure along all three cross-sections, is ideal for bringing out the effects, if any, of sample cross-section on the estimated "*f*" values by different methods of measurements. The estimated "*f*" parameters are listed in Table 2a. Similarly, Table 2b and c list the values of "*f*" parameter estimated from all the methods in case of "Zry-2 Rod" and "Hot-ex-Zr-2.5Nb" samples. Following important observations can be made from the "*f*" parameter values listed in these tables for the three samples considered in this study.

- For 'Rex-Zry-4' and 'Zry-2 Rod', "*f*" values determined by EBSD were nearly identical in all the cross-sections. For 'Hot-ex-Zr-2.5Nb', however, small but noticeable deviations in "*f*" values were observed among the cross-sections. It needs to be noted that this sample, generalized in Table 1 as partially recrystallized, had lower success in automated EBSD indexing.
- PF estimated f_{RD} values in ND–TD were always inconsistent with corresponding values from other cross-sections.
- Degree of scatter in "*f*" parameters determined by XRD methods, across different cross sections is in general more than that of the EBSD.

Fig. 4 brought out another aspect of comparison. As shown in the figure, qualitative trend in average "*f*" was similar between the techniques. This was achieved by neglecting measurements from ND–TD cross-sections in case of the PF technique and also by normalizing the IPF estimated "*f*" parameters.

EBSD is expected to provide direct estimates of area/volume fractions of texture components. This is unlike the indirect measurements through XRD, where estimations depend on the detector efficiency and, in case of the ODF method, on the adopted numerical method. The main problem of EBSD is in statistics – i.e. lower statistics in terms of limited area of scans and also lower indexing in regions of high defect density. In case of 'Rex-Zry-4' and 'Zry-2 Rod' samples, relatively larger scan area (2 mm × 2 mm) and the fact that the samples were fully recrystallized addressed these issues quite successfully. Nearly identical "*f*" values, at three different cross-sections, certify this apparent success. However, in case of 'Hot-ex-Zr-2.5Nb' the partially recrystallized nature of the microstructure led to lower success in indexing. This led to noticeable scatter in the estimated "*f*" values along different cross sections. In addition, widely different grain sizes along different cross sections of this sample result in wide variation in the effective number of grains covered per given area of measurement. This highlights the variability introduced into the texture measurement by the variation in microstructure. In spite of these limitations, EBSD estimated values were comparable with other techniques, Table 2a–c.

As was mentioned, the degree of scatter in "*f*" parameter across different cross sections was more for the X-ray based methods (see Table 2a–c). PF method strongly depended on the availability of sufficiently strong (0 0 0 2) peak. (0 0 0 2) intensity in the ND–TD cross section was negligible: an example is shown in Fig. 2 for 'Rex-Zry-4'. The same is expected in all Zr-based components with strong basal texture. In all such cases, the ND–TD cross section needs to be avoided for the PF method. This also explains inconsistent values of PF estimated f_{RD} from ND–TD cross section. Though PF estimated "*f*" parameters between the other two cross sections (RD–TD and RD–ND) differed as well, their average was in close agreement with that of the EBSD values. This can be attributed to missing of certain poles at high tilt angles during PF measurement, particularly beyond 75° (in fact this tilt corresponds to a grazing angle of 15° in the X-ray set up used in the present study). Measurements at high tilt angles are typically neglected in the so-called reflection based pole figure measurements because of high defocusing of X-ray. This issue does not exist in transmission based technique at these high tilt angles. The transmission based technique, on the other hand, has serious problems of sample preparation and also of defocusing at the low tilts [3]. Mainly because of the difficulties in sample preparation, from late '1950s X-ray pole figure measurements almost became routine through the reflection technique [3]. In any such measurement, however, data from high tilts (typically $\psi \geqslant 75°$, as in the present study) are neglected. Of course this creates problems in "*f*" parameter estimations. In principle, "*f*" parameter estimation requires covering full 90° of tilt in any given cross section, which is experimentally not possible using the conventional reflection technique alone. This creates an inherent problem in any conventional pole figure based *f* parameter estimation from single cross section. When measurements are performed on different cross sections, which are at 90° tilts,

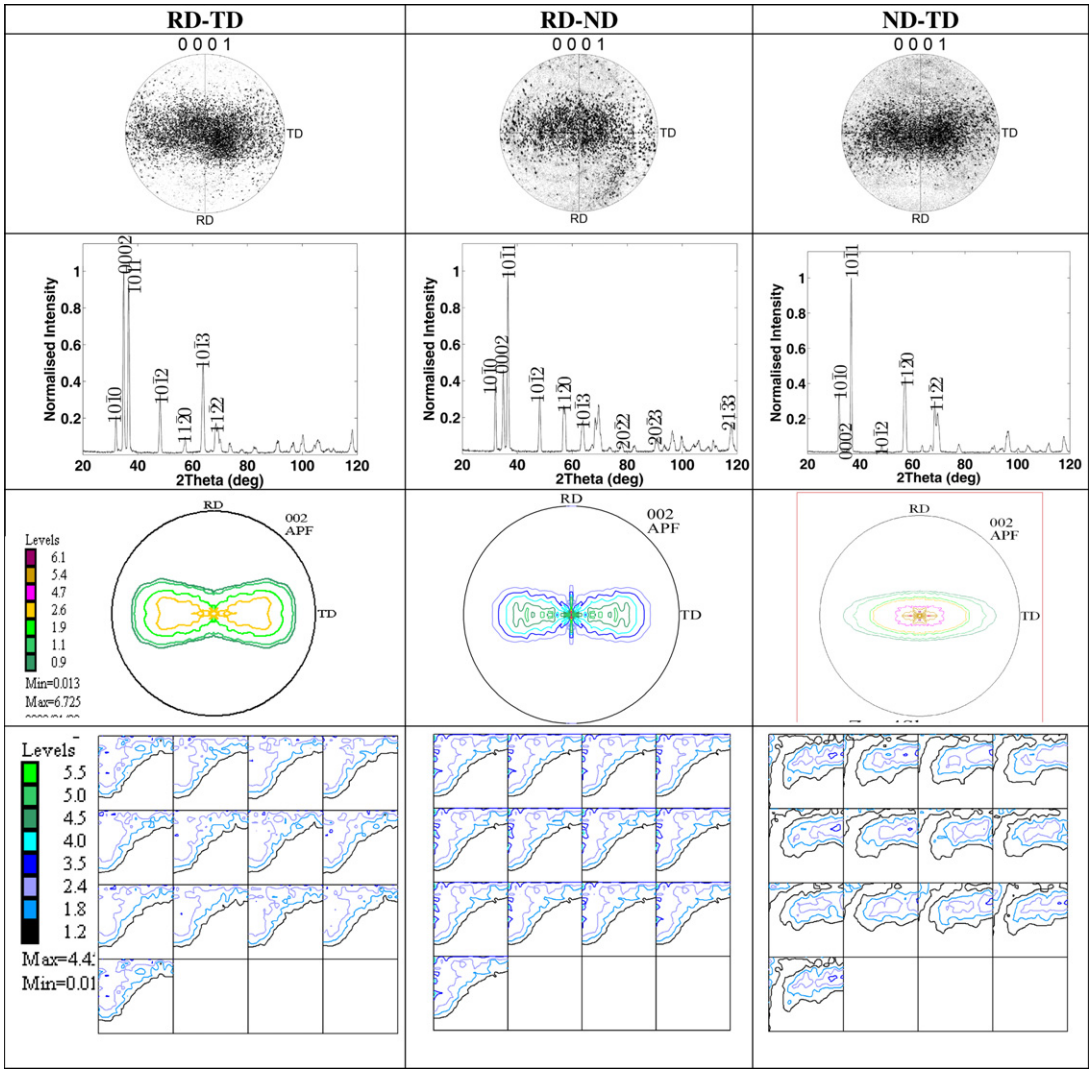

Fig. 3. Comparison of EBSD estimated discrete pole figures, X-ray diffractograms, XRD estimated pole figures and ODFs - as determined along the three different principal cross sections i.e. RD–TD, RD–ND, ND–TD for the case of "Rex-Zry-4". The 0002 PFs along the RD–ND and ND–TD cross sections are derived from corresponding ODF. The ODF in the RD–TD and RD–ND were determined by using $(0\,0\,0\,2)$, $(01\bar{1}2)$, $01\bar{1}3$ and $(02\bar{2}2)$ while the ODF of ND–TD cross section was determined using $(10\bar{1}0)$, $(01\bar{1}2)$, $(01\bar{1}3)$ and $(02\bar{2}2)$.

Table 2
F parameters of samples estimated by different methods.

	EBSD			IPF		PF			ODF		
F	Plane of measurement			As calculated	Normalized	Plane of measurement			Plane of measurement		
	RD–TD	RD–ND	ND–TD			RD–TD	RD–ND	ND–TD	RD–TD	RD–ND	ND–TD
(a) Rex-Zry-4											
F_{ND}	0.31	0.32	0.32	0.36	0.35	0.23	0.38	0.26	0.30	0.35	0.38
F_{RD}	0.10	0.10	0.10	0.16	0.15	0.12	0.10	0.25	0.18	0.10	0.11
F_{TD}	0.58	0.58	0.58	0.52	0.50	0.66	0.52	0.49	0.51	0.54	0.50
(b) Zry-2 Rod											
F_{ND}	0.44		0.41	0.60	0.47	0.32		0.31	0.44		0.43
F_{RD}	0.01		0.05	0.09	0.07	0.06		0.28	0.09		0.08
F_{TD}	0.45		0.44	0.60	0.47	0.66		0.40	0.47		0.47
(c) Hot-ex-Zr-2.5Nb											
F_{ND}	0.43	0.42	0.32	0.43	0.35	0.412	0.31	0.36	0.43	0.31	0.41
F_{RD}	0.07	0.09	0.03	0.11	0.09	0.05	0.06	0.24	0.10	0.21	0.13
F_{TD}	0.50	0.49	0.65	0.68	0.56	0.54	0.63	0.39	0.46	0.48	0.45

averaging such measurements appeared to offer better estimations of "*f*" parameters – see Fig. 4. Strictly speaking, contribution of basal intensity from all of the three cross sections should be taken into account, to fill the entire angular range needed for "*f*" parameter calculation. However, the error introduced by neglecting a given cross section will be minimum if the basal intensity from that

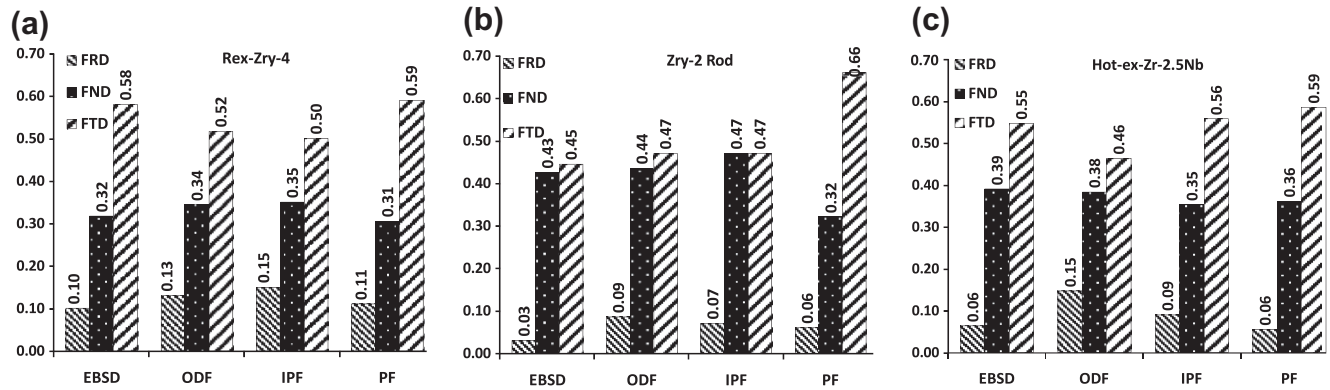

Fig. 4. Comparison of the F_{RD}, F_{ND} and F_{TD} (average values from the respective sections) determined by different methods: (a) "Rex-Zry-4", (b) "Zry-2 Rod" and (c) "Hot-ex-Zr-2.5Nb". In case of PF method, values from ND–TD cross section were ignored in calculating the average values. The IPF values in the figure are normalized ones (see Table 2).

cross section is negligible. This is the reason why averaging from two cross sections (RD–TD, RD–ND) with high basal intensities has given rise to consistent values of "*f*" parameter, in spite of neglecting the contribution from ND–TD section (where basal intensity is negligible, see Fig. 3).

IPF estimated "*f*" parameters did not sum up to 1.0, explainable from the assumption of linear variation of diffraction intensities across different rotation angles [4]. As noted originally by Kearns [5], the angular positions of the diffraction peaks used for the "*f*" parameter determination, are not distributed uniformly on the IPF space and hence, averaging of the intensities over a given tilt angles can lead to slight errors. However, the normalized values were in good agreement with the EBSD values. Normalization appears to have rounded off the errors committed in approximations introduced in this method. The main advantage of this technique is that this method does not require a texture stage. Although this is a simple technique to apply, requirement of having all the three cross sections for the determination of the "*f*" parameter places additional constrains on the suitability of this method. If the samples are very thin in one dimension (as in case of clad tubes), generating sample along ND–TD or RD–ND direction involves stacking together of several slices and necessitates appropriate additional corrections in the measured intensities. In addition, for a non homogeneous material, since different sample regions are scanned in arriving at the "*f*" values, possibilities of deviations are also large.

ODF estimated f_{RD} in the ND–TD cross section appeared more realistic than the same values estimated by PF technique in the same cross section. This was achieved by selecting an alternate non-basal pole and subsequent ODF calculations. Basal fiber volume fractions were thus estimated from ODF and were not as sensitive to the actual existence of the (0 0 0 2) peak.

Present study thus, brought out that the EBSD method provided the most consistent "*f*" parameter estimation, particularly when the microstructure is well recrystallized (low defect/dislocation density) and sufficient area of the sample is scanned to improve the statistics. Traditionally X-ray based methods were in more use for the determination of the *f* parameter as they are comparatively faster, and can handle the highly deformed microstructures (with high defect/dislocation densities). However, as demonstrated in the present study, availability of the faster cameras made EBSD technique comparable to the X-ray based techniques both in time of measurement and in statistics with the added advantage of improved consistency. EBSD method did not, however, offer any advantage over X-ray methods in case of deformed microstructures. In case of deformed samples, the quality of the EBSD pattern and hence the success of the method are severely affected by the presence of crystalline defects such as dislocations. X-ray based methods can be more convenient to apply in these cases, as they are insensitive to the extent of sample deformation. Among the different X-ray based methods, ODF method, contrary to popular belief, did not offer any significant improvement in the accuracy of *f* parameter determination. However this method suffered less from lack of sufficient basal pole intensity in the measurement cross section in comparison to the corresponding PF method. Normalizing in case of IPF method and taking average from two suitable sections in case of PF method did improve the accuracy of *f* parameter estimations. Based on these observations, it is recommended to use EBSD technique particularly in case of recrystallized samples and either IPF or PF method in case of highly deformed materials.

4. Conclusion

Present study showed that the EBSD method is the most consistent for the determination of the "*f*" parameter provided that sufficiently large area is scanned with high success rate of indexing, as in case of recrystallized microstructures. In case of partially recrystallized microstructures, however, all the methods considered in the present study did show scatter in the "*f*" values across different cross sections. The X-ray based methods of IPF, PF and finally the ODF followed the EBSD in terms of consistency of the estimated "*f*" values. The importance of selection of appropriate section was brought out in case of PF method. It was also shown that the suitability of method depends also on the measurement cross section and type of texture.

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